



सेंद्रल ट्रान्समिशन यटिलिटी ऑफ इंडिया लिमिटेड

(पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड के स्वामित्व में)

(भारत सरकार का उद्यम)

CENTRAL TRANSMISSION UTILITY OF INDIA LTD.

(A wholly owned subsidiary of Power Grid Corporation of India Limited)

(A Government of India Enterprise)

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Subject: 26th Consultation Meeting for Evolving Transmission Schemes in Northern Region-Minutes of Meeting

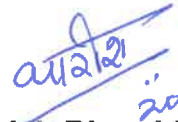
Dear Sir/Ma'am,

Please find enclosed the minutes of the 26th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 1st December, 2023 (Friday) through virtual mode.

The minutes are also available at CTU website (www.ctuil.in)

Thanking you,

Yours faithfully,


20/12/23

(Kashish Bhambhani)
General Manager (CTU)

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Minutes for 26th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 01.12.2023**A. Application related matters in Northern Region****A1. Applications for Connectivity to ISTS under CERC (Connectivity and General Network Access to the inter-State Transmission System) Regulations, 2022 received in Oct'23:**

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)
1.	2200000254	Avaada RJBikaner Private Limited	Bikaner distt., Rajasthan	03.10.2023	Generator (Solar)	Land Route	600	31.03.2025	Bikaner-III PS
2.	2200000275	Renew Solar (Shakti Six) Private Limited	Jodhpur distt., Rajasthan	05.10.2023	Generator (Solar)	Land Route	400	30.06.2026	Bhadla-III PS
3.	2200000281	Anboto Solar Private Limited	Barmer distt., Rajasthan	06.10.2023	Generator (Solar)	Land BG Route	250	30.11.2025	Barmer PS
4.	2200000266	Malur Renewables Private Limited	Bikaner distt., Rajasthan	06.10.2023	Renewable Power Park Developer (Solar)	Land Route	650	30.05.2025	Bhadla-III PS
5.	2200000300 (Enhancement)	Solarcraft Power India 4 Private Limited	Jaisalmer distt., Rajasthan	08.10.2023	Generator (Solar)	Land Route	50	16.06.2025	Bhadla-III PS
6.	2200000305	Juniper Green Energy Private Limited	Barmer distt., Rajasthan	11.10.2023	Generator (Solar)	Land BG Route	300	31.03.2027	Barmer-I PS

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)
7.	2200000289	Avaada Energy Private Limited	Bikaner distt., Rajasthan	12.10.2023	Generator (Solar)	Land BG Route	300	30.11.2025	Bhadla-III PS
8.	2200000309	Renew Solar (Shakti Six) Private Limited	Jodhpur & Bikaner distt., Rajasthan	13.10.2023	Generator (Solar)	Land Route	585	30.06.2026	Bhadla-III PS
9.	2200000284	AK Renewable Infra Private Limited	Bikaner distt., Rajasthan	13.10.2023	Generator (Solar)	Land Route	300	30.06.2025	Bikaner-I PS
10.	2200000312	Auxo Sunlight Private Limited	Barmer distt., Rajasthan	14.10.2023	Generator (Solar)	Land BG Route	300	30.09.2026	Barmer-I PS
11.	2200000290 (Enhancement)	Avaada Energy Private Limited	Barmer distt., Rajasthan	17.10.2023	Generator (Hybrid)	Land BG Route	250 (Solar-125 Wind-125)	30.09.2025	Fatehgarh-IV PS
12.	2200000316	Renew Sun Power Private Limited	Barmer & Jaisalmer distt., Rajasthan	17.10.2023	Generator (Solar)	Land BG Route	300	30.09.2026	Barmer-I PS
13.	2200000327	Hazel Hybren Private Limited	Jaisalmer distt., Rajasthan	20.10.2023	Generator (Solar)	Land BG Route	300	31.01.2026	Ramgarh PS
14.	2200000306	Shudh Solar Power Private Limited	Bikaner distt., Rajasthan	21.10.2023	Generator (Solar)	Land Route	250	31.05.2025	Bikaner-III PS
15.	2200000308	NTPC Renewable Energy Limited	Bikaner distt.,	21.10.2023	Generator (Solar)	Land Route	700	31.03.2026	Bikaner-III PS

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)
			Rajasthan						
16.	2200000334	Hazel Hybren Private Limited	Bikaner distt., Rajasthan	23.10.2023	Generator (Solar)	Land BG Route	300	31.01.2026	Bikaner-III PS
17.	2200000319	AM Green Energy Private Limited	Bikaner distt., Rajasthan	23.10.2023	Generator (Solar)	Land BG Route	300	31.05.2025	Bikaner-II PS
18.	2200000333	Furies Solren Private Limited	Jaisalmer distt., Rajasthan	27.10.2023	Generator (Solar)	Land BG Route	300	31.01.2026	Ramgarh PS
19.	2200000355	Gamma Renewables India Project One Private Limited	Barmer distt., Rajasthan	31.10.2023	Renewable Power Park Developer (Solar)	Land BG Route	300	01.06.2026	Barmer-I PS

It was deliberated that considering application received in Oct'23, no further margin available at Bhadla-III PS and Bikaner-IV PS (upto 3.6GW). In view of above transmission scheme for Bhadla-IV PS and Bikaner-IV PS (beyond 3.6GW) is under planning as some of applications received in Oct'23 month may also be granted with transmission system under planning.

Further, Transmission scheme for 5.5GW at Fatehgarh-IV (Sec-2)/Barmer-I PS is already under bidding, however with new applications received exceeding 5.5GW would need additional transmission scheme which is under planning. It is to mention that in Rajasthan, application for about 58 GW capacity connectivity is already granted/discussed. Further, as discussed in the meeting held under chairmanship of Hon'ble Minister of Power & NRE on 01.11.23, due to concentration of projects in particular locations in Rajasthan, power evacuation is becoming difficult and therefore going forward this may require HVDC corridors/ lines which required more time for execution. Therefore, based on evolution of such transmission scheme, above applications (Oct'23) will be deliberated in CMETS-NR meeting (Dec'23) along with planned transmission schemes.

A2. Application for Grant of GNA/GNA_{RE} to entities other than STU under Regulation 20.1, 20.3 and 20.4 to entities under Regulation 17.1(ii), (iii), (v) and (vi):

SI No.	Application ID	Name of the Applicant	Submission Date	Region	Nature of applicant	GNA within Region (MW)	GNA outside Region (MW)	Total Quantum (MW) of GNA Required	Start date of GNA	End date of GNA
1	2200000272	HHP Two Private Limited	06.10.2023	NR	Bulk consumer seeking to connect to ISTS at Sonipat (PG)	0.0	110	110	01.05.2026	01.05.2051

It was stated that application from M/s HHP Two Private Limited Bulk consumer for seeking connectivity to ISTS (GNA of 110MW -outside the region) will also be deliberated in next CMETS-NR meeting as system studies are undergoing to assess the requirement of transmission system strengthening at Sonipat(PG), if any.

B. Proposal of revised dedicated Transmission system for Connectivity to M/s Adani Renewable Energy Holding Sixteen Limited (180 MW) & Adani Renewable Energy Holding Four Limited(500 MW) at Fatehgarh-II PS

It was stated that Adani Renewable Energy Holding Sixteen Limited(AREH16L)(erstwhile SBE Renewables Sixteen Private Limited)(St-II App no. 1200002450 :180 MW) & Adani Renewable Energy Holding Four Limited(AREH4L)(St-II App No. 1200002430: 500 MW) had been granted Stage-II connectivity at Fatehgarh-II PS.

The above entities are also granted LTA for the same capacity. Transition of connectivity of above entities under GNA Regulations have been discussed and agreed in the 22nd CMETS-NR meeting held on 21.08.2023. Subsequently, CTU vide letter dated 25.09.2023 granted deemed GNA to above entities under 37.3 of GNA Regulations.

However, M/s Adani vide letter dated 04.10.2023 & 21.11.2023 requested for revision of connectivity transmission system. M/s Adani informed that, in accordance with the instructions provided by the GIB committee and in order to optimize

transmission corridor, they have proposed a hybrid arrangement, combining overhead and underground transmission lines, connecting the Solar Plants to Fatehgarh-II PS as below:

1. Adani Renewable Energy Holding Sixteen Limited(180 MW):

The connectivity transmission system earlier granted to M/s AREH16L (erstwhile SBE Renewables Sixteen Private Limited):

- 1) SBE Renewables Sixteen Private Limited Power Plant – Fatehgarh-II PS 220kV S/c line (suitable to carry min 300MW power at nominal voltage) along with associated bay at generation end: under scope of the applicant
- 2) 220kV Bay at Fatehgarh-II PS shall be under the scope of ISTS.

The proposed revised connectivity scheme by M/s Adani:

- 1) Adani Renewable Energy Holding Sixteen Limited Solar Power Plant PS at Devikot – Common point: 220 kV underground cable- approx.-5 kms -**Under Applicant scope**
- 2) 220 kV Single circuit line on Multicircuit Tower sharing with existing Transmission line of Adani Renewable Energy Holding Nineteen Private Limited (AREH19PL) from common point to Fatehgarh-II PS covering approx.-10 kms.-**Under Applicant scope**
- 3) 220kV Bay at Fatehgarh-II PS shall be under the scope of ISTS.

2. Adani Renewable Energy Holding Four Limited(500 MW):

The connectivity transmission system earlier granted to M/s AREH4PL:

- 1) Adani Green Energy Four Ltd. Power plant – Fatehgarh-II PS 220 kV D/c line (suitable to carry min 300 MW power per circuit under nominal voltage) along with associated bay at generation end: under scope of applicant.
- 2) 220 kV Bays at Fatehgarh-II PS shall be under the scope of ISTS.

The proposed revised connectivity scheme by M/s Adani:

- 1) Adani Renewable Energy Holding Four Limited Solar Power Plant PS at Bhimsar – Common point: 220 kV D/c line on D/c tower- approx.-8 kms. -**Under Applicant scope**
- 2) 220 kV underground Cable from common point to Fatehgarh-II PS – approx.-8 kms. -**Under Applicant scope**
- 3) 220 kV Bays at Fatehgarh-II PS shall be under the scope of ISTS.

The revised proposals as above (1 & 2) for Adani Renewable Energy Holding Sixteen Limited (180 MW) & Adani Renewable Energy Holding Four Limited(500 MW) were agreed in the meeting.

All statutory clearances in respect of above DTL configuration shall be the responsibility of M/s Adani.

C. ISTS Expansion

i. Transmission system scheme for Ratle HEP (850MW)

It was deliberated that in 3rd Consultation meeting for Evolving Transmission Schemes in NR held on 28/01/2022, connectivity system for Ratle HEP (850MW) was deliberated. In the meeting, It was informed that the transmission system for evacuation of Ratle HEP (850 MW) was discussed and agreed during the 4th NRPC(TP) meeting held on 05/10/2021 & 12/10/2021 wherein it was deliberated that Ratle HEP (850 MW) is a Run of River scheme located on river Chenab, in Kishtwar District of Jammu & Kashmir, with design overload of 10% and its completion schedule is Feb/March 2026. It was also agreed that Ratle HEP is approx. 20 km from Kishtwar PS, therefore, it can be directly connected with Kishtwar PS with 400 kV D/c line with high capacity conductor. During the above meeting, it was deliberated that RHPCL shall explore the possibility of single-phase units of either 125 MVAR bus reactor or 80 MVAR reactor in case of transportation constraints.

In this meeting, RHPCL informed that they have already awarded 1x125 MVAR Bus Reactor at generation switchyard, wherein, supply of 1-phase or 3-phase units is left to the EPC contractor's scope. Accordingly in 3rd CMETS-NR meeting, it was decided to grant connectivity to RHPCL for Ratle HEP (850 MW) through following transmission system for Connectivity:

- Ratle HEP - Kishtwar (GIS) PS 400 kV D/c line (line suitable for carrying around 935 MW on each circuit at nominal voltage) along with 2 nos. of 400kV bays at Kishtwar S/s - To be implemented by the applicant (RHPCL)
- 125 MVAR, 420 kV bus reactor at Ratle HEP - To be implemented by applicant (RHPCL)
- Establishment of 400 kV Kishtwar (GIS) PS by LILO of one circuit of Kishenpur – Dulhasti 400kV D/c (Quad) line (Single Circuit Strung) and 125 MVAR Bus reactor at Kishtwar (GIS) PS– being implemented under ISTS

Further in 23rd CMETS-NR meeting held on 29.08.23, application for GNA transition of M/s RHPCL (Applicant) under 37.2 of GNA regulation was deliberated with following details

S. No	Name of Applicant	Connectivity Quantum (MW)	System for GNA	Start Date Requested	Start date of Connectivity/GNA	Bank Guarantee Requirement as per GNA Regulation (BGs, if any submitted in Connectivity Regulations, 2009 shall be adjusted)
1	Ratle Hydroelectric Power Corporation Limited	850	Augmentation Without ATS	16-Feb-26	16-Feb-26 (Interim)	Conn-BG1 : Rs. 50 lakh Conn-BG3 : Rs. 2 lakh/MW

In the meeting it was stated that confirmation was taken from CEA for applicability of hydro potential zones for Ratle HEP. CEA confirmed that Ratle HEP falls under hydro potential area in HP/J&K. Considering this, the transmission system for Connectivity under GNA was proposed as augmentation without considering Associated Transmission System (ATS) as given below:

Details of Associated Transmission System (Augmentation Without ATS):

A. Associated Transmission System (ATS): NIL

B. Transmission system for Connectivity (under scope of Applicant):

- 400 kV Ratle HEP Switchyard - Kishtwar D/c line along with bays at both ends

C. Common Transmission system for Connectivity under GNA (under ISTS):

- 400 kV Kishenpur-Samba D/c line(Quad) (2nd)
- Bypassing of one ckt of 400kV Kishtwar – Kishenpur D/c line (Quad) at Kishenpur and connecting it with one of the circuit of Samba-Kishenpur 400kV D/c line(Quad), thus forming 400kV Kishtwar-Samba (Quad) direct line (one ckt)
- Termination of another circuit of 400 kV Kishenpur-Samba D/c line(Quad) at Kishenpur at 400Kv bay vacated at Kishenpur (After bypassing of one ckt of 400 kV Kishtwar-Kishenpur at Kishenpur)
- Bypassing both ckts of 400kV Kishenpur – Samba D/c line (Twin) & 400 kV Samba – Jalandhar D/c line (Twin) at Samba and connecting them to form Kishenpur– Jalandhar D/c direct line (Twin)
- 400kV Samba- Jalandhar D/c line(Quad) (only one circuit is to be terminated at Jalandhar while second circuit would be connected to bypassed circuit of Jalandhar –Nakodar 400kV D/c line)
- Bypassing 400kV Jalandhar – Nakodar line (Quad) at Jalandhar and connecting it with one of the circuit of Samba-Jalandhar 400kV D/c line(Quad Moose), thus forming Samba –Nakodar line
- Termination of another circuit of 400 kV Samba -Jalandhar D/c line (Quad) at Jalandhar at bay vacated at Jalandhar (After bypassing of one ckt of 400 kV Jalandhar - Nakodar at Jalandhar)
- Reconductoring of remaining portion of Dulhasti-Kishtwar 400 kV D/c line with Quad conductor
- Reconductoring of 400 kV Kishenpur-Dulhasti S/c line (Twin) (minimum 2100 MVA capacity)

Further, in the meeting it was deliberated that the above scheme is tentative, which shall be finalized in the ensuing CMETS-NR meeting. The scheme along with implementation schedule shall be confirmed as part of in principle grant of connectivity. Feasibility of HTLS conductor has been reviewed in view of altitude aspects. It was also deliberated during the meeting that the above-mentioned CON-BG1 & 3 shall need to be submitted within two months from intimations. As part of scheme, Reconductoring of remaining portion of Dulhasti-Kishtwar 400 kV D/c line (twin moose) and Reconductoring of 400 kV Kishenpur-Dulhasti line (twin zebra) is also required.

In 3rd NRPC-TP meeting it was also deliberated that outlet beyond Dulhasti is Dulhasti-Kishenpur 400kV line which is a single circuit line, the amount of power that can be exported/imported is limited. Hence, Dulhasti - Ratle section would be optimized to the extent possible. bay rating at Dulhasti is 2000A, which is further reduced to 700/800A due to reduced capacity of XLPE/OIL cable for connection of line to GIS bus at Dulhasti end.

In view of above a meeting was conveyed on 13.09.23 by CEA and CTU with NHPC to discuss upon feasibility for upgradation of switchgear equipment's incl. cable at 400kV Dulhasti switchyard as well as reconductoring of part system of 400kV Dulhasti-Kishtwar line. In the meeting, it was decided that NHPC shall explore the feasibility for upgradation of the switchgear and cables at Dulhasti substation to rating of 3150 A. NHPC may also explore the option of replacement of cables with GIB bus duct. The same would be intimated to CEA within one month.

NHPC vide mail dated 16.10.23 informed that GIS bay equipment installed at Dulhasti Power Station have a rating of 2000A. As suggested, complete replacement of GIS system would be required to accommodate requisite 3150 A. Accordingly, the existing rating of GIS busbar and associated switchgear like CB, CT, ES/ISO etc. needs to be uprated to 3150 A from 2000A. The compatibility issue of New GIS with old GIS will attract huge financial implication. It is envisaged that the space for the complete GIS compatible for 3150A may be required more than that of existing one and may not be feasible to comfortably accommodate within the existing underground space. Further, the existing items of pothead yard like wave traps, Isolators etc. also needs to be replaced which has an additional cost implication.

Further NHPC informed that there are two 400 kV cables (Line-1 is oil filled and Line-2 XLPE insulation) of 630 Sq.mm each are used for Power evacuation from Dulhasti GIS which can carry approx. 700-800A. As suggested, the replacement of cables with GIB bus duct may not be feasible at Dulhasti Power Station considering the constraints of cable tunnel space.

Considering the above facts, the uprating of all the above items will require significant Prolonged outages of generating units causing Generation and PAF Loss which will attract huge financial loss on account of loss of generation. Further, equipment replacement cost would also put an additional financial implication. In view of above factors, it is not possible to uprate the existing systems at Dulhasti Power station from 2000 A to 3150 A.

In view of above constraints, various other options were carried out and studies were reviewed. In the revised proposal, LILO of 400 kV Kishenpur- Dulhasti line (2nd ckt) at Kishtwar S/s (one ckt already LILOed as part of Pakaldul transmission system) along with Reconductoring of 400 kV Kishenpur-Kishtwar line (formed after LILO of Kishenpur- Dulhasti line at Kishtwar S/s) is proposed. In the revised scheme, there will be no requirement of space for installation of new GIS equipment and replacement of cables at Dulhasti switchyard for time being. Study files were also circulated on 24.11.23.

In the CMETS meeting it was stated that as part of transmission scheme, 400kV Kishenpur-Samba D/c line(Quad) line is proposed. Due to higher loading on 400/220kV ICTs as well as space constraint at Kishenpur S/s, bypassing of 400kV Kishtwar – Kishenpur line (Quad) at Kishenpur and connecting it with one of the circuit of Kishenpur-Samba 400kV D/c line (Quad) (thus forming 400kV Kishtwar-Samba (Quad) direct line (one ckt) & Kishenpur- Samba (Quad) (other ckt) line) is proposed. Further 4 nos. of bays shall be vacated at Samba S/s through bypassing both ckts of 400kV Kishenpur – Samba D/c line (Twin) & 400 kV Samba – Jalandhar D/c line (Twin) at Samba and connecting them together to form Kishenpur– Jalandhar D/c direct line (Twin). These bays will be utilized in present scheme through Kishenpur-Samba D/c line(Quad) line (one ckt bypassed at Kishenpur) & 400kV Samba- Jalandhar D/c line(Quad) line.

In view of space constraint for 400kV bays at Jalandhar S/s as well as higher loading on 400/220kV ICTs, bypassing 400kV Jalandhar – Nakodar line (Quad) at Jalandhar and connecting it with one of the circuit of Samba-Jalandhar 400kV D/c line (Quad) is proposed (forming Samba –Nakodar direct line). The vacated bay at Jalandhar S/s will be utilized for termination of another circuit of 400 kV Samba -Jalandhar D/c line (Quad).

It was deliberated that CTU vide mail 20.11.23 to POWERGRID requested information regarding maximum rating of the HTLS conductor suitable for Reconductoring of 400 kV Kishenpur-Kishtwar line (Twin Zebra), considering altitude and the bundle size of the line. It was also requested that POWERGRID may confirm the feasibility of the line bay upgradation requirements at Kishenpur Substation pertaining to the above line. In this regard POWERGRID vide mail 08.12.23 informed that 400 kV Kishenpur-Dulhasti line (Twin Zebra) line has bay equipment rating of 2000A (CB,CT, Isolator). In view of that bays upgradation is also required with Reconductoring of 400 kV Kishenpur-Kishtwar line (Zebra). Conductor configuration for reconductoring of 400 kV Kishenpur-Kishtwar line will be finalized based on POWERGRID inputs.

In the latest JCC meeting, M/s RHPCL informed commission schedule for their 850MW (4x205+1x30MW) HEP project. progressively from May'26. Further in the present meeting, M/s RHPCL confirmed the commissioning schedule of generation project progressively from May'26. It was also requested that M/s RHPCL may also confirm generation schedule through mail/letter.

Grid-India in the meeting as well as in mail 04.12.23 queried about expected timelines of other HEP in the area such as Kiru, Kavar, Kirthai-I, Kirthai-II etc. as maximum of nearly 4000MW could be evacuated through 3 outgoing lines from Kishtwar i.e. 2 lines to Kishenpur and one line to Samba.

CTUIL informed that at present connectivity application received at Reoli Dugli, Purthi, Bardang and Dugar HEPs with schedule progressively from 2029, however no application is yet to received from Kiru, Kavar, Kirthai-I and Kirthai-II HEPs.

For evacuation of power from Reoli Dugli, Purthi and Bardang HEPs, separate high capacity transmission system will be planned and based on studies, connectivity of above high capacity corridor with Kishtwar S/s (via Dugar HEP) will be finalized.

Grid-India also stated that fault level at Kishtwar & Kishenpur also need to be checked after the proposed interconnections. Based on the fault level assessments & transmission plan, switchgear rating may be selected. From the studies it is emerged that with Ratle HEP, fault level of Kishtwar S/s is about 23kA (designed capacity:50KA) and Kishenpur S/s is about 37kA (designed capacity: 40kA) which is within limits.

Further Grid-India stated that the requirement of line reactors for lines such as 400kV Kishtwar-Kishenpur (132km) and 400kV Kishtwar-Samba (160km) may be studied given the persistent issues of high voltage in hydro complex in winter months during off-peak hours. CTU informed that requirement of line reactor will be reviewed considering space availability. Accordingly, CTU analyzed the reactive compensation requirement, however feasibility for installation of line reactors along with space confirmation from POWERGRID & Sterlite is awaited. Same will be intimated in next CMETS-NR meeting.

Grid-India also mentioned in mail that the bay allocation of feeders at Kishtwar station is very important & incomer and outgoing feeders need to be kept in same diameter. PSTCL stated that they are planning to establish new substation at 400kV Wadala Granthian for which LILO of 400kV Jalandhar-Samba is under planning. CTU stated that at present with proposed scheme 400kV Samba-Jalandhar (Twin) (after bypassing above line will be Kishenpur-Samba (Twin)) and 400kV Samba-Jalandhar (Quad) line will be available for LILO in 2026-27 timeframe. In view of that PSTCL may carry out the studies for both the alternatives and share with CEA & CTU. PSTCL agreed for the same.

In view of above, following ISTS transmission scheme is agreed for evacuation of power for Ratle HEP (850MW):

Common Transmission system for Connectivity under GNA (under ISTS) :

1. 400 kV Kishenpur-Samba D/c line (Quad) (2nd) - 35kms
2. Bypassing of 400kV Kishtwar – Kishenpur line (Quad) (Under implementation as part of Pakaldul transmission scheme) at Kishenpur and connecting it with one of the circuit of Kishenpur-Samba 400kV D/c line(Quad), thus forming 400kV Kishtwar-Samba (Quad) direct line (one ckt)
3. Termination of another circuit of 400 kV Kishenpur-Samba D/c line(Quad) at Kishenpur at 400kV bay vacated at Kishenpur (After bypassing of one ckt of 400 kV Kishtwar-Kishenpur at Kishenpur) –160km
4. Bypassing both ckts of 400kV Kishenpur – Samba D/c line (Twin) & 400 kV Samba – Jalandhar D/c line (Twin) at Samba and connecting them together to form Kishenpur– Jalandhar D/c direct line (Twin)

5. 400kV Samba- Jalandhar D/c line(Quad) (only one circuit is to be terminated at Jalandhar while second circuit would be connected to bypassed circuit of Jalandhar –Nakodar 400kV D/c line) – 205km
6. Bypassing 400kV Jalandhar – Nakodar line (Quad) at Jalandhar and connecting it with one of the circuit of Samba-Jalandhar 400kV D/c line(Quad Moose), thus forming Samba –Nakodar line
7. Termination of another circuit of 400 kV Samba -Jalandhar D/c line (Quad) at Jalandhar at bay vacated at Jalandhar (After bypassing of one ckt of 400 kV Jalandhar - Nakodar at Jalandhar)-177km
8. LILO of 400 kV Kishenpur- Dulhasti line (Twin Zebra) at Kishtwar S/s (one ckt already LILOed as part of Pakaldul transmission system)
9. Reconductoring of 400 kV Kishenpur-Kishtwar line* (Twin HTLS) (minimum 2100 MVA capacity) (formed after LILO of Kishenpur- Dulhasti line at Kishtwar S/s) along with bay upgradation works (2000A to 3150A) at Kishenpur end for above line

*** Conductor configuration for reconductoring of 400 kV Kishenpur-Kishtwar line will be finalized based on POWERGRID inputs. Line reactors will be finalized based on POWERGRID/Sterlite inputs**

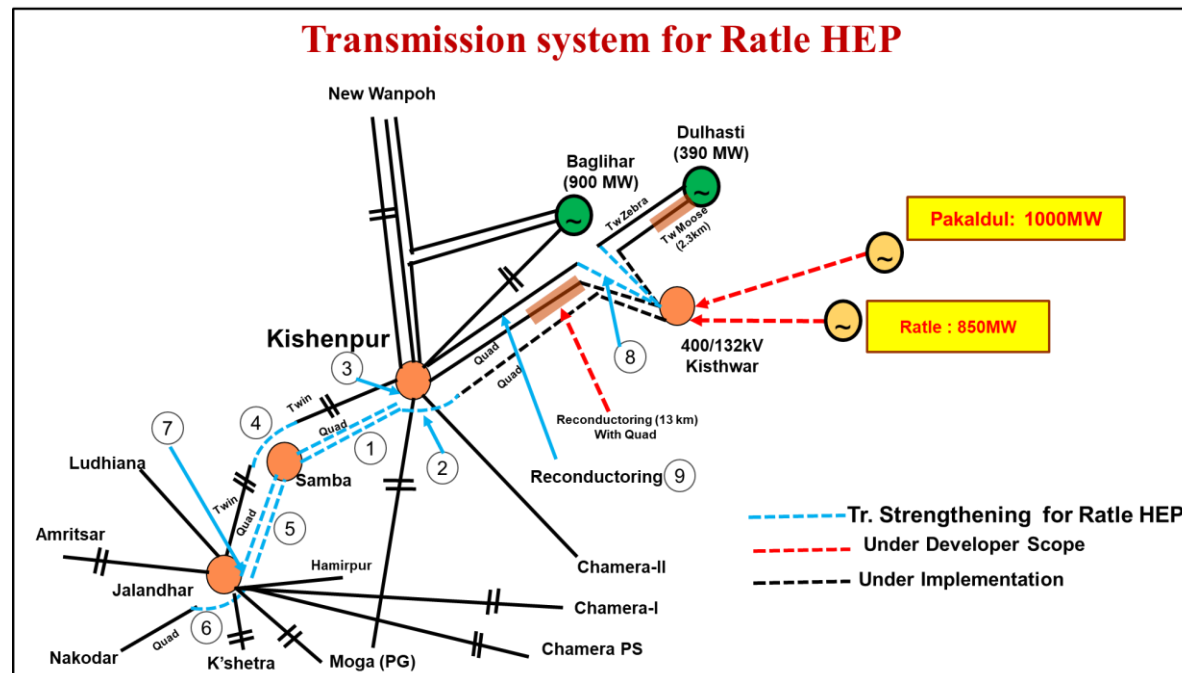


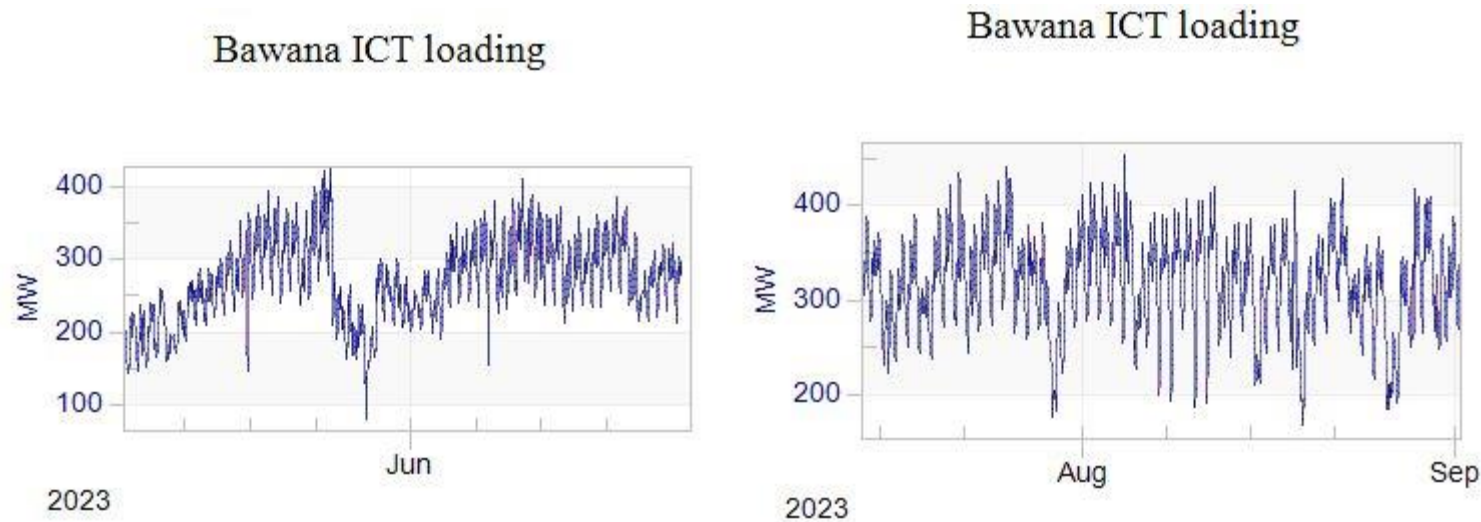
Fig: Transmission scheme for Ratle HEP (Scheme S.No. as mentioned above is indicated near each element in diagram)

ii. **Replacement of 1x315 MVA (ICT-3), 400/220 kV ICT to 500 MVA at 400/220 kV Bawana S/s**

It was stated that CEA vide mail 30.08.23 informed that DTL vide its letter 28.08.23 to CEA, submitted the proposal for replacement of existing 400/220 kV, 315 MVA ICT (ICT-3, POWERGRID Asset) at Bawana substation with 500MVA ICT, to meet the peak load requirement of Delhi. In the letter, DTL has informed that there are 6 nos. of 400/220 kV, 315 MVA ICTs installed at Bawana substation of DTL out of which ICT-3 belongs to POWERGRID and rest 5 nos. ICTs belong to DTL. Due to high fault level, bus split operation has been implemented at Bawana and 2x315 MVA ICT (ICT-2 &3) ICT, 4x315 MVA ICT (ICT-1,4,5 & 6) are kept on different bus sections. As per loading conditions at Bawana substation, DTL has already planned replacement of ICT-2 and ICT-4 with 500 MVA ICTs. In addition, replacement of ICT-3, which belongs to POWERGRID is also required to make the system 'N-1' compliant. In view of that, CEA requested to CTU & Grid-India to furnish comments on the proposal.

GRID INDIA vide mail dated 01.09.23 informed that due to split bus arrangement, the section having 2x315MVA ICTs is N-1 non-compliant at the time of high demand of Delhi state control area. The issue of N-1 non-compliance is being continuously raised by NRLDC in NR-OCC and NRPC meetings and also submitted in quarterly operational feedback to CTUIL/CEA. On highlight of the issue by NRLDC, DTL has implemented SPS at Bawana S/s section having 2x315 MVA ICTs. Considering above, Grid-India suggested to go ahead with ICT augmentation at 400/220kV Bawana with the future plan of drawl by Delhi from Bawana and nearby area to ensure N-1 compliance on 2 ICT section(s).

CTU vide mail 06.09.23 informed that from the real time loading pattern shared by Grid-India, it emerged that total loading on 2x315 MVA ICTs (in one section) are in order of 400-450MW during peak load scenario and ICTs may become N-1 non compliant in near future with growing demand. Loading pattern of Bawana ICTs is as under **(Source-Grid-India)**



Loading of Bawana ICTs (ICT-2 & 3 : 2x315MVA)

Therefore, replacement of both the ICTs (ICT-2 & 3) is required to ensure N-1 compliance. CTU also stated that establishment of 400/220kV Tikrikhurd Substation along with LILO of 400kV Bawana-Mandola D/c line were earlier approved in 2nd Meeting of NRPC-TP held on 1.09.2020, however the scheme is not yet awarded by DTL. It is requested that DTL may expedite the implementation of Tikrikhurd Substation as it will relieve the loading on Bawana ICTs in future. As per discussion with DTL, ICT-3 (315MVA) along with bays is owned by POWERGRID as part of ISTS scheme. In view of that the proposal need to be taken up in ensuing CMETS-NR meeting.

CEA vide letter dated 18.09.23 informed that replacement of both the 400/220 kV, 315 MVA ICTs (ICT-2 & 3) with 500 MVA ICTs is required to ensure 'N-1' compliance, out of which, replacement of ICT-2 is already planned by DTL. In view of the above, proposal submitted by DTL for replacement of 400/220 kV, 315 MVA ICT (ICT-3), is generally in order.

CTU vide mail 11.09.23 requested POWERGRID to provide information regarding year of make for ICT-3 (315MVA) and, its remaining useful life (ICT & associated bays) as well as healthiness of above transformer (Residual Life Assessment Studies if any).

POWERGRID vide email dated 18.09.2023 and letter dated 03.10.23 informed that 400/220/33 kV, 315 MVA ICT-3 along with associated bays (owned by POWERGRID) at Bawana substation was commissioned in June'1997. The equipment of associated bays of said ICT at Bawana have been replaced under ADD CAP (CB, CT & relays). Remaining bay equipment i.e. Isolators, CTs, Surge Arrestors are being replaced under ADD CAP 2019-2024. It is further to mention that ICT (BHEL make) which was commissioned in Jun'1997 was failed in Sep'18 and new ICT (Siemens make, YO - 2011) was commissioned in Nov'2018 in place of failed BHEL make ICT. The present ICT is in healthy condition and may be kept as regional spare. Grid-India stated that POWERGRID may inform OCC for above 315MVA ICT to be used as regional spare.

Grid-India & CEA in the meeting agreed for the proposal and emphasized on urgent requirement of replacement of both the ICTs (ICT-2 in Intra state and ICT-3 in Inter state) at Bawana S/s. DTL agreed for the requirement of ICT-3 (ISTS). CEA and CTU also stated that DTL may also expedite the replacement of ICT-2 (315 MVA with 500MVA), for N-1 compliance at Bawana S/s. DTL agreed for the same.

In view of above deliberations, following ICT augmentation scheme is agreed in ISTS:

- Replacement of 400/220/33 kV, 1x315 MVA (3rd) ICT (ICT-3) to 500 MVA at 400/220 kV Bawana (DTL) S/s

iii. Transmission system for evacuation of power from Bikaner Complex as part of Rajasthan REZ Ph-IV(Part-3 :3.6GW)

It was stated that transmission system for evacuation of power from Bikaner Complex as part of Rajasthan REZ Ph-IV (Part-3 :3.6GW) was agreed in 25th CMETS-NR meeting held on 31.10.23. In the above meeting it was requested by CTU for confirmation of any power drawl requirement from proposed Churu & Siwani S/s in future by RVPN & HVPN respectively.

Subsequently, HVPN vide letter dated 20.11.23 informed that there will be no requirement of HVPN for drawl of power (at 220kV) in future from proposed Siwani S/s. RVPN vide letter dated 21.11.23 informed that in the present scenario neither 220kv bays nor 400/220kV ICTs are required by RVPN at proposed 765/400kV Churu S/s. In the meeting, CTU asked SECI to confirm RE potential at Churu and Siwani. SECI informed that they will intimate the information regarding RE potential at Churu and Siwani S/s within 2-3 days. It was decided that based on SECI inputs, decision on deletion of 220kV scope at Churu and Siwani will be taken.

SECI vide mail 07.12.23 informed that there is no viable Wind resource potential in Churu or Siwani S/s, however South West Churu has solar potential in the area as indicated below.

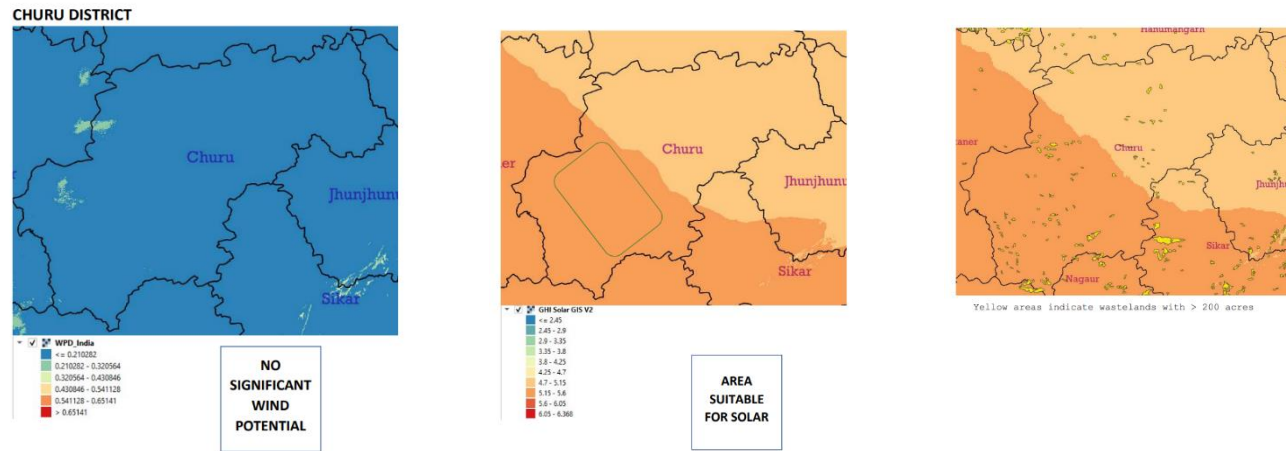


Fig: WPD and Solar (GIS) maps provided by SECI

Considering solar potential at Churu as indicated by SECI, 220kV future scope shall be retained at Churu S/s as earlier agreed in 25th CMETS-NR meeting. However, future 220kV scope (400/220kV ICTs along with bays -4 nos., 220 kV line bays for drawl -4 nos. & 220kV Sectionalization bay: 2 sets) shall be deleted from Siwani S/s in Rajasthan REZ Ph-IV (Part-3 :3.6GW) scheme.

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